「Def.

Let $E \subset X$ be a non-empty subset of a Metric space X. Define

$$\rho_E : X \to \mathbb{R} : x \mapsto \inf_{z \in E} d(x, z)$$

is called a distance from x to E.

「Prop. Let ρ_E be a distance from x to E in a Metric space X.

1). $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

2). ρ_E is Continuous Uniformly.

Proof. To show the first, Using the Property of closure and infimum;

$$x \in \overline{E} \iff \forall \varepsilon > 0,\ B_\varepsilon(x) \cap E \neq \emptyset \iff \forall \varepsilon > 0,\ \exists y \in E \text{ such that } d(x,y) < \varepsilon \iff \inf_{z \in E} d(x,z) = 0.$$

To show the second assertion, observe that: given $x, y \in E$,

$$\rho_E(x) \leq d(x,z) \leq d(x,y) + d(y,z) \text{ for any } z \in E$$

Since it holds for all $z \in E$,

$$\rho_E(x) \leq d(x,y) + \rho_E(y)$$

and which implies $\rho_E(x) - \rho_E(y) \leq d(x,y)$. Similarly, $\rho_E(y) - \rho_E(x) \leq d(x,y)$, So

$$|\rho_E(x) - \rho_E(y)| \leq d(x,y) \text{ for any } x, y \in X.$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$, then we obtain that

$$d(x,y) < \delta \Rightarrow |\rho_E(x) - \rho_E(y)| \leq d(x,y) < \varepsilon \qquad \boxtimes$$

⌈Lemma. Let $A, B \subseteq X$ be non-empty disjoint closed set in Metric space X. Then, a map $f : X \to \mathbb{R} ; x \mapsto \dfrac{\rho_A(x)}{\rho_A(x) + \rho_B(x)}$ where $\rho_E : X \to \mathbb{R} ; x \mapsto \inf_{z \in E} d(x, z)$ is continuous function satisfying $A = f^{-1}[\{0\}]$ and $B = f^{-1}[\{1\}]$.

Proof. At first, since

$$\rho_A(x) + \rho_B(x) = 0 \iff \rho_A(x) = \rho_B(x) = 0 \iff x \in \overline{A} \cap \overline{B} = A \cap B = \emptyset,$$

so $\rho_A(x) + \rho_B(x) \neq 0$ for all $x \in X$, so f is well-defined.
And, since ρ_A and ρ_B are continuous, $\rho_A + \rho_B$ also continuous, so $f = \rho_A / (\rho_A + \rho_B)$ is continuous, being rational of continuous function.⌋

Observe that: Since $\rho_A + \rho_B > 0$ and $\rho_A, \rho_B \geq 0$,

$f(x) = 0 \iff \rho_A(x) = 0 \iff x \in \overline{A} = A$ and

$f(x) = 1 \iff \rho_B(x) = 0 \iff x \in \overline{B} = B$, therefore

$$A = f^{-1}[\{0\}] \text{ and } B = f^{-1}[\{1\}]. \qquad \boxtimes$$